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Abstract
Purchasing a home is typically the largest financial investment most people ever make. With ever-growing costs, potential buyers scour sites in the hopes of gaining positive returns. However, most commercial home purchasing platforms only focus on the positive aspects of parcels for sale, while omitting negatives and nonconformative user preference inputs. We introduce Land Alchemist - the data and visualization platform for analyzing how features external to a home affects its value and total cost of ownership over time. The platform allows home buyers and researchers to input preferences, simulate scenarios to explore changes in relationships, and make better-informed decisions.

How is it done today; what are the limits of current practice?
Currently, tools and studies exist to see how amenities serve us when buying in an area, but each focuses on one or a few features. Landfills, toxic plants, power plants and contamination lower nearby home prices [1, 3, 5, 6, 12], while schools, colleges, hospitals, rail and everyday amenities raise them [2, 4, 7, 11], but the effects differ by location and user [2,10]. Also they do not let the user interactively see how new amenities or disamenities such as datacenters could change the property's value. Existing tools map amenities without negatives, forecasts [9], or predict prices without letting users weigh features [8].

What's new in your approach? Why will it be successful? Expected Innovations
Land Alchemist incorporates a dynamic research sandbox where users have freedom to adjust sensitivities based on personal preferences and circumstances. Evidence shows that homes and people respond to various features differently depending on preference or capabilities [10]. Each feature's sensitivity is a parameter in an algorithm the user can change with a slider, and the map and charts update the new forecasted price. In addition to being dynamic, our approach includes multiple negative features which are typically scrutinized in isolation. These capabilities will be displayed in one platform allowing for comparison of multiple zip codes on one screen creating a new way to visualize valuation variances. The combination offers users a novel research sandbox.

Who cares?
Preference plays a factor in pricing for both buyers and sellers. Out-of-state buyers are less informed about the area as well [6]. Land Alchemist allows for all wanted features to be seen in one place for buyer ease. Lenders will also benefit as they can minimize risk to exposures based on their preference. Evidence shows that this is common practice, where lenders decide to deny loans when homes are exposed to disamenities [10]. Finally, local governments care as property prices shape their tax base. Evidence shows that the presence of unwanted features in an area could lead to declined property values and thus lower tax revenues generated by municipalities [6]. This tool allows for officials to input current perspectives on certain features to dynamically price in tax shortfalls or windfalls. Overall, according to evidence it is seen as a social benefit to incorporate a formal cost analysis [5].

If you're successful, what difference and impact will it make, and how do you measure?
More informed buyers and sellers leads to a more efficient marketplace. If Land Alchemist is successful, it will help both parties in price discovery on transactions that are in the magnitudes of hundreds of thousands to millions of dollars.

Measuring whether the project achieves its purpose can be done through user testing by real estate agents, buyers, and researchers to use the project and rate their experience with a survey. This approach is similar to other projects with similar tooling that used Likert scale surveys [9].

What are the risks and payoffs?

Technical risks: Some features, like datacenters, are new and have little data, so their predicted effects could be mostly inaccurate. Most effects only reach a limited distance and change from place to place [2], but our data is by ZIP code, so small effects may get lost. Utility risks: It's possible Land Alchemist is not viewed as helpful by users and the researchers would be out the opportunity cost of time to develop the platform. Payoffs: One fixed price does not fit every buyer. Through better discovery of prices based on user preferences, users can approach opportunities with calculated prices that fit their personalized evaluation. Land Alchemist could reinvent the evaluation experience.

How much will it cost?

We will not have a cost to the project. We will use free educational digital resources and publicly available data sets.

What are the midterm and final exams? Plan of Activities

Activity	Milestone	Date Range	Team Members	Gantt Chart
Proposal Documentation	-	10/2 - 10/15	Andrei, Anik, Nick, Tharun	
Explored data sets for 10 features	Midterm	10/9 - 11/6	Andrei, Tharun	
Harvested predicted value data sets	Midterm	10/15 - 11/6	Anik, Tharun	
Create mock ups of visualizations, discuss user journey	Midterm	10/25 - 11/6	Andrei, Nick	
Identified multiple features to predicted value sensitivities	Midterm	10/25 - 11/6	Anik, Tharun	
Create basic structure of app (backend/frontend)	Midterm	10/28 - 11/4	Andrei, Nick	
Document build instructions	-	11/1 - 11/4	Andrei, Anik, Nick, Tharun	
Data lake export POC and documentation	-	11/5 - 11/7	Nick	
Data cleaning and joining	-	11/5 - 11/6	Andrei, Anik, Nick, Tharun	
Analysis to determine static weights	-	11/5 - 11/6	Nick	
Determine formula for dynamic weights	-	11/5 - 11/9	Andrei, Anik, Nick, Tharun	
Application Testing	Final	11/9 - 12/1	Andrei, Anik, Nick, Tharun	
Application supports user workflows, user interactivity, and produces visualizations	Final	11/1 - 12/4	Andrei, Anik, Nick, Tharun	
User testing and feedback	Final	12/1 - 12/4	Nick	
Poster Preparation	Final	12/1 - 12/4	Andrei, Anik, Nick, Tharun	

All team members have contributed a similar amount of effort

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